



SAC

2025

André Pereira

The new manycore generations
(most slides & images are borrowed)

From multicore to manycore: key issues



Tips to fit many cores into a single chip

- lower the compute capability of each core, but not too much
- use a scalable interconnection network fabric on-chip (NoC)
 - to minimize shared-cache/memory access latencies
 - to provide enough data communication bandwidth
 - to minimize traffic bottlenecks
- group cores in clusters to improve the quality of the NoC
- reduced cache size/levels (*beware: strong impact on performance...*)
- smaller fabrication process (Sapphire Rapids: 7nm; Zen 4: 5 nm; Apple M3: 3nm)
- mix general-purpose PUs with application-oriented modules:
GPUs for vector computing, TPU for tensor computing, ...
- move to MCM/chiplets (*simpler chips have better wafer production yield*)

Interconnect Fundamentals

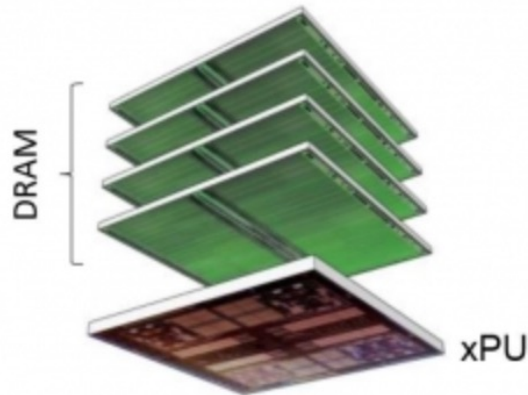


- **Networks-on-chip**: an adapted follow-up of interconnection systems to link servers in supercomputers
- Key parameters that define a **NoC**:
 - **topology**: defines how the nodes and links are connected, namely all possible paths a message can take through the network
 - **routing algorithm**: selects the specific path a message will take from source to destination
 - **flow control protocol**: determines how a message actually traverses the assigned route
 - **router micro architecture**: implements the routing and flow control protocols and critically shapes its circuits

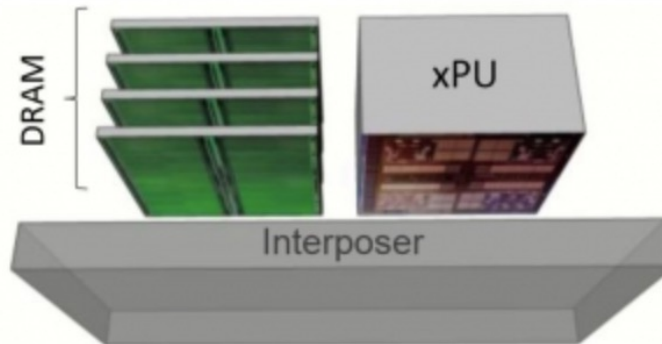
3D Stacked DRAM

BUT FIRST... THERE ARE TWO FLAVORS OF DIE STACKING

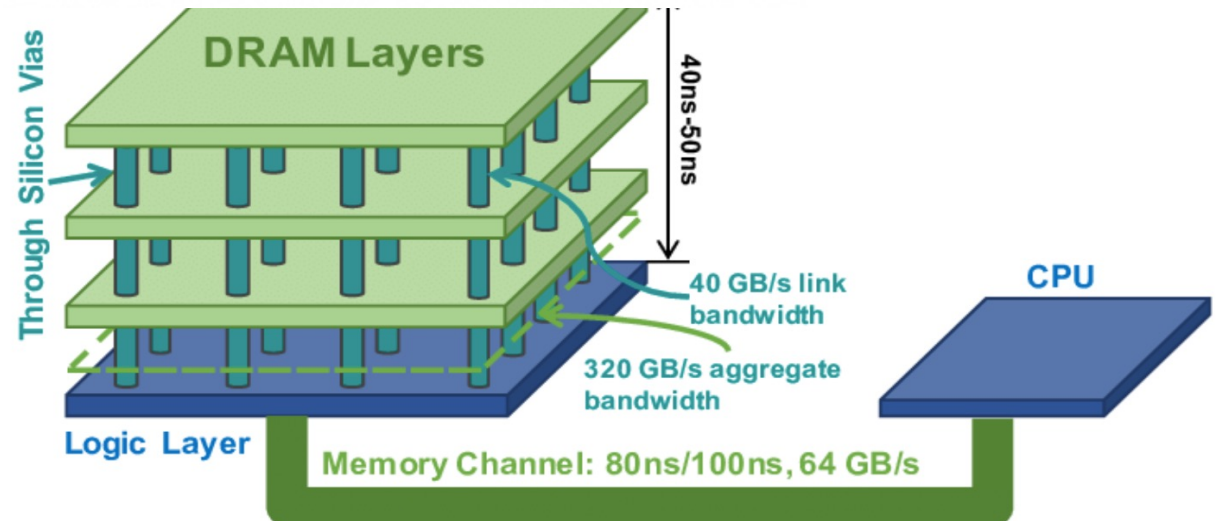
AMD



VERTICAL STACKING (3D)



INTERPOSER STACKING (2.5D)



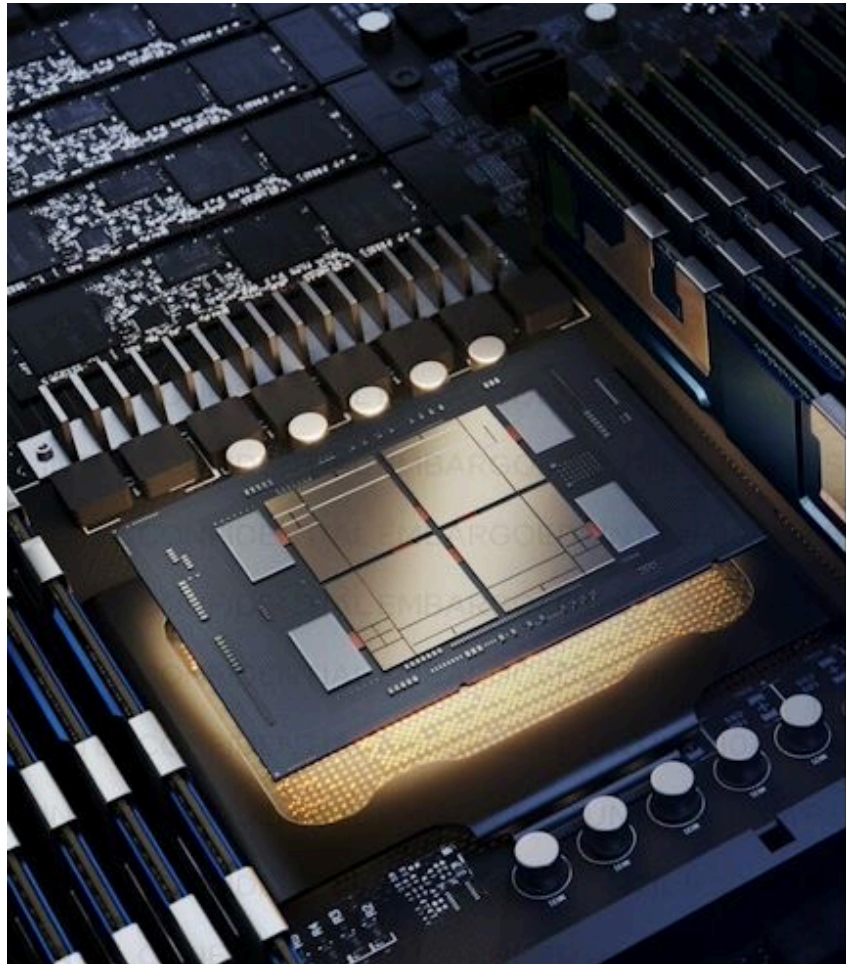
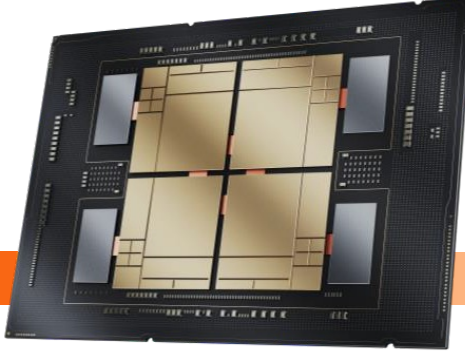
Manycore chips/packages: an overview on current servers



Key server chips/packages that addresses those issues:

- **Intel: the Xeon Scalable family**
- AMD: the Epyc Zen family
- ARM: key ARMv8 & v9 server-level competitors
 - Fujitsu A64FX Arm chip
 - Neoverse hyperscale reference design for Amazon Graviton
 - Alibaba Yitian 710
 - Huawei HiSilicon Kunpeng 920
- Sunway: the SX260x0 family
- Cerebras: a Wafer Scale Engine
- Apple (*not server...*): the SoC approach (*no chiplets!*)

4th Gen Intel Xeon Max Family: Sapphire Rapids (Q1' 23)

A photograph of an Intel Xeon Max processor mounted on a server motherboard. The processor is a large, square, gold-colored chip with a complex circuitry pattern. It is surrounded by various components like capacitors and other chips on the dark green PCB. The image is part of a promotional graphic for the Sapphire Rapids HBM processor.

Intel® Xeon® processors codenamed

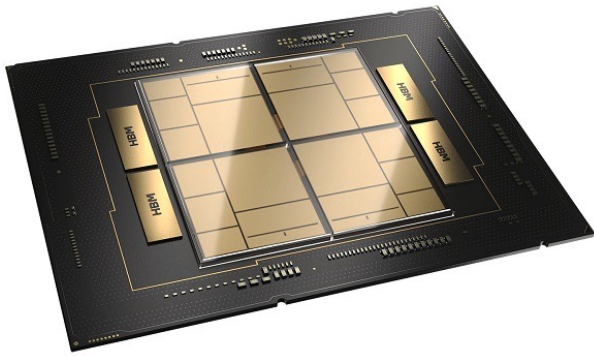
Sapphire Rapids

HBM

AMX
Advanced Matrix Extensions

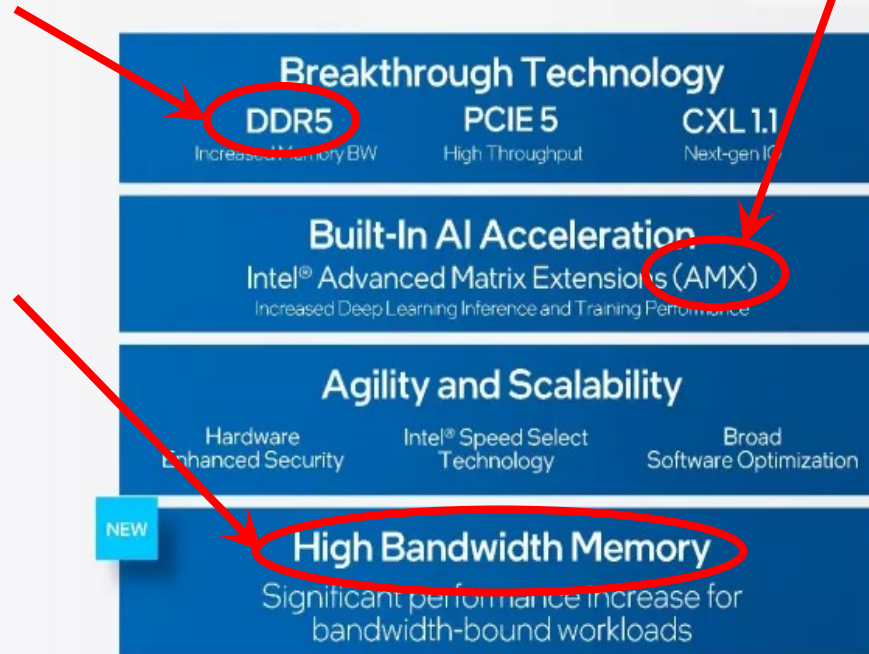
Integrated Acceleration Engines

64GB
HBM2e



4th Gen Intel Xeon Max Family: Sapphire Rapids (Q1'23)

Next-Generation Intel Xeon Scalable Processors
Unique Capabilities Optimized for HPC and AI Acceleration



Up to
64 GB
HBM2e

4x 14-core tiles
4x 2 mem ch

intel.

Manycore chips/packages: an overview on current servers

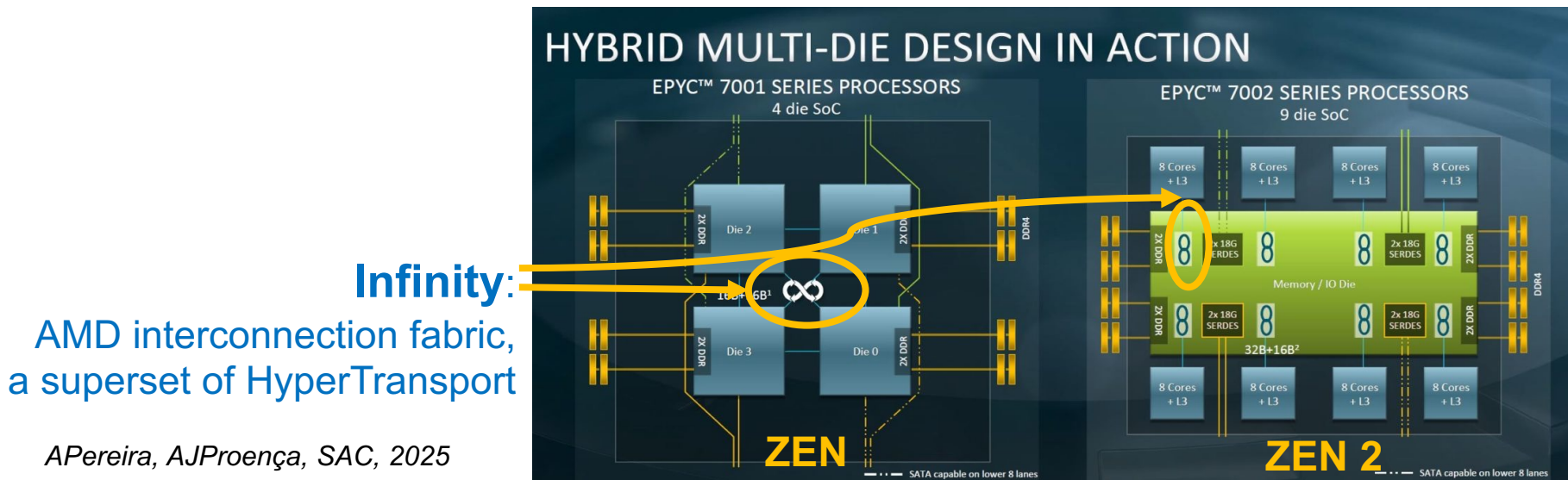
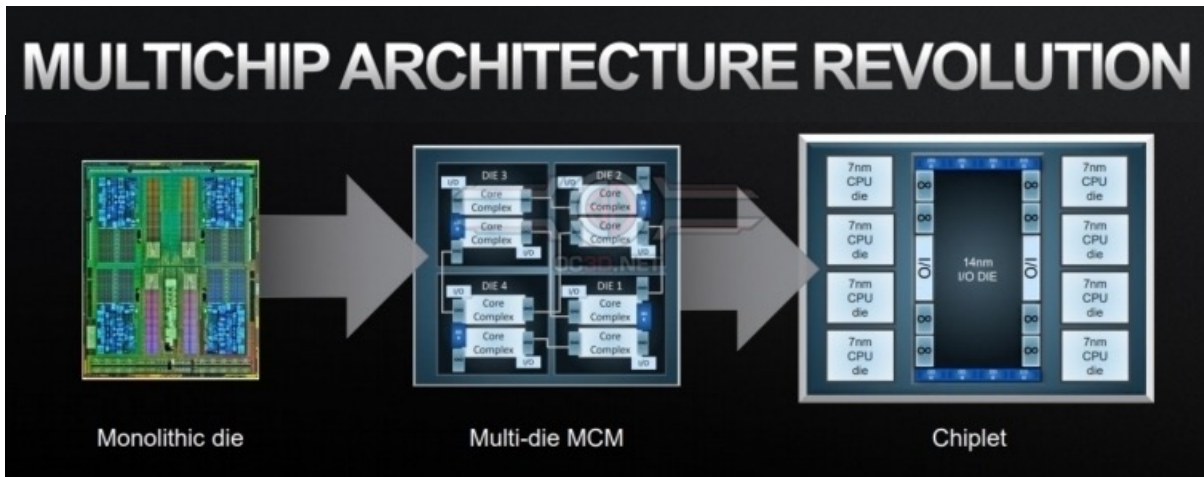


Key server chips/packages that addresses those issues:

- Intel: the Xeon Scalable family
- **AMD: the Epyc Zen family**
- ARM: key ARMv8 & v9 server-level competitors
 - Fujitsu A64FX Arm chip
 - Neoverse hyperscale reference design for Amazon Graviton
 - Alibaba Yitian 710
 - Huawei HiSilicon Kunpeng 920
- Sunway: the SX260x0 family
- Cerebras: a Wafer Scale Engine
- Apple (*not server...*): the SoC approach (*no chipelets!*)



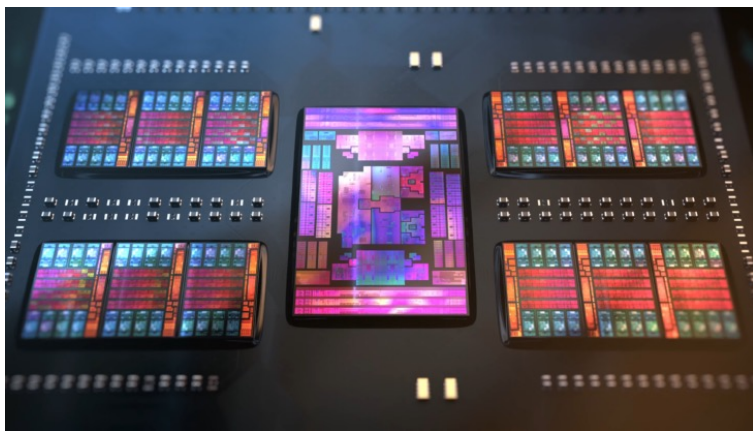
Key Intel Xeon competitor: AMD Epyc (Zen, Zen 2, 3, 4)





<https://wccfttech.com/amd-epyc-rome-7nm-server-cpu-official-launch-64-core-128-thread-128-pcie-gen4/>



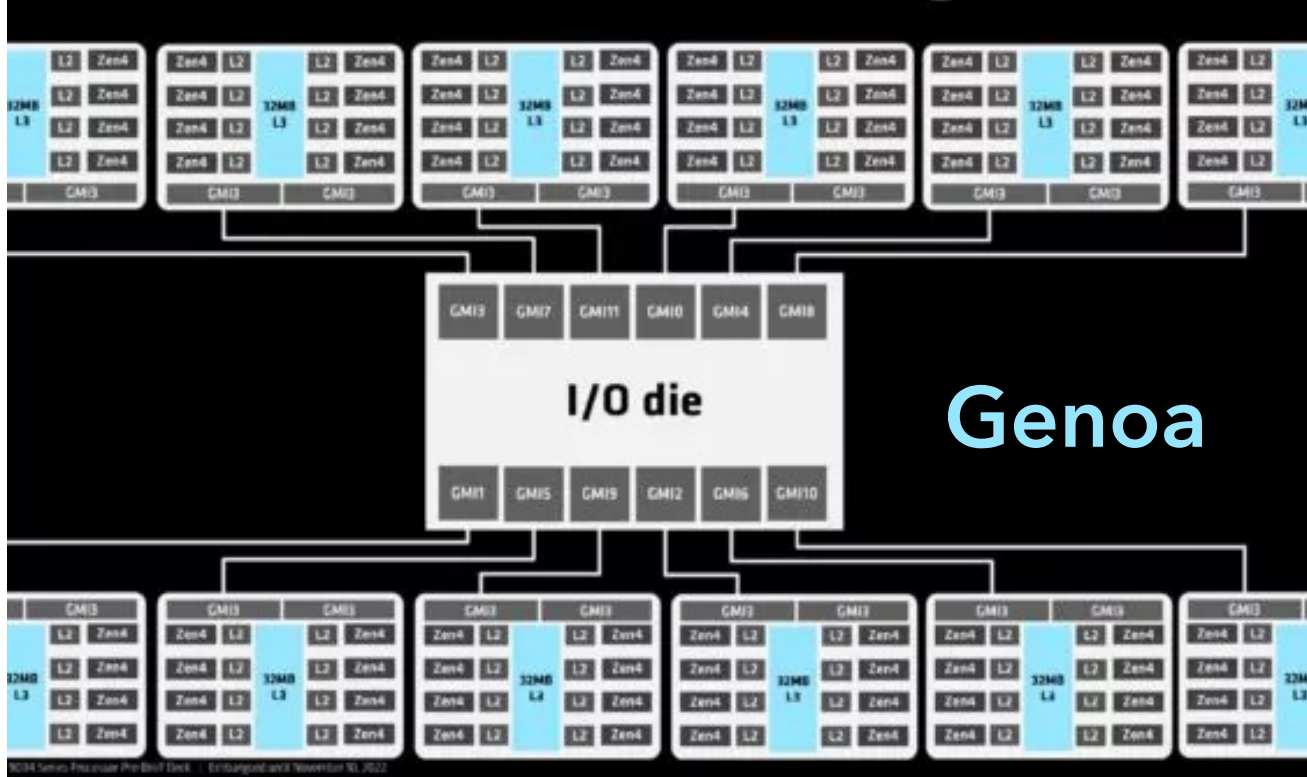


AMD 4th Gen Epyc: Zen 4 (Genoa & Bergamo)

Genoa: up to **5nm 96 cores** /192 threads
BFLOAT16, VNNU, AVX-512

1 MiB/core L2 cache
384 MiB L3 cache
12 **DDR5** mem chan
128 PCIe Gen 5.0
(launched Nov'22)

AMD EPYC™ 9004: 12 CCD Configuration



Genoa

Bergamo:

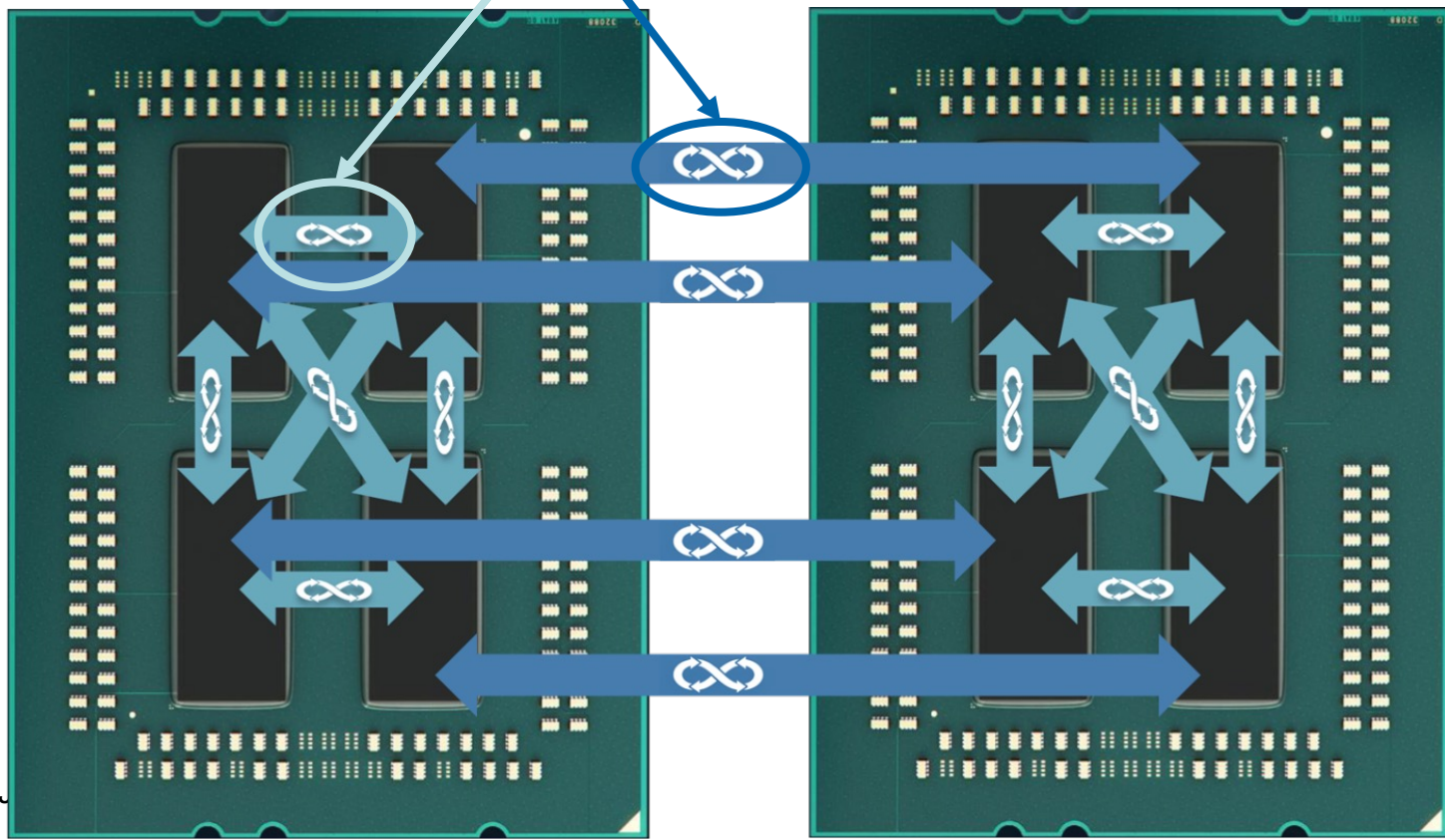
- up to **128 Zen 4c** cores
- for cloud-native computing
(expected 2H 2023)



The AMD Infinity Fabric

Infinity Fabric (IF)

- AMD system interconnect architecture, a 256-wide bi-directional crossbar:
 - **Infinity Fabric On-Package (IFOP):** die-to-die communication in same package
 - **Infinity Fabric InterSocket (IFIS):** for package-to-package communications



Manycore chips/packages: an overview on current servers

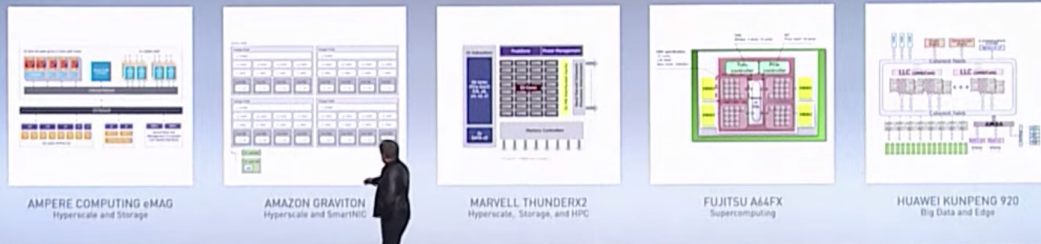


Key server chips/packages that addresses those issues:

- Intel: the Xeon Scalable family
- AMD: the Epyc Zen family
- **ARM: key ARMv8 & v9 server-level competitors**
 - Fujitsu A64FX Arm chip
 - Neoverse reference design for Amazon Graviton
 - Alibaba Yitian 710
 - Huawei HiSilicon Kunpeng 920
- Sunway: the SX260x0 family
- Cerebras: a Wafer Scale Engine
- Apple (*not server*): the SoC approach (*no chiplets!*)

Support for
at least dual-socket

HPCs with ARMv8: server-level competitors (back in 2019...)



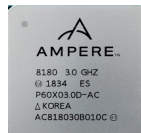
1. Marvell ThunderX product family (**dead?...**)



2. Fujitsu A64FX Arm chip (**in MACC...**)



3. Neoverse N1 hyperscale reference design (...)



4. Ampere Altra Arm Processor (...)



5. Amazon Graviton (...)



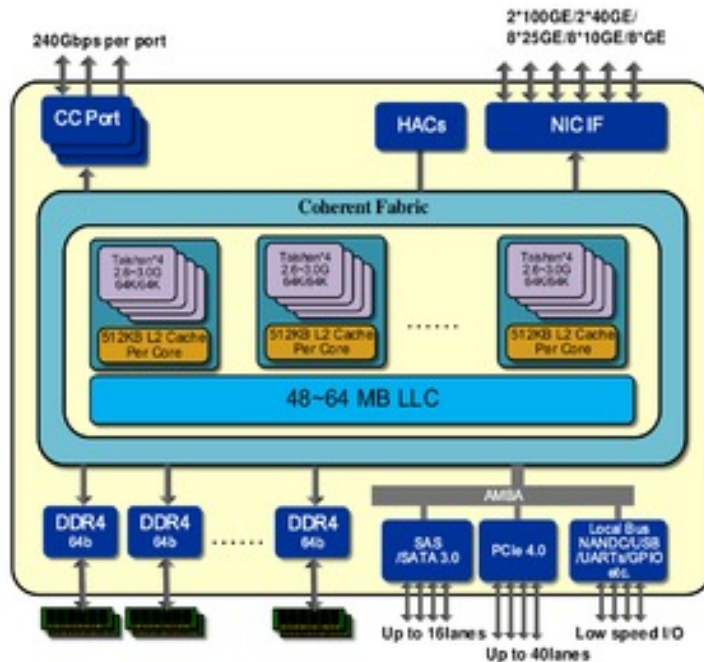
6. Huawei HiSilicon Kunpeng 920 (**dead?...**)

The Huawei Kunpeng 920 (previously known as HiSilicon Hi1620)

(launched in 2019)

Huawei Kunpeng 920: based on TaiShan V110 core, a semi-custom ARM Cortex-A72

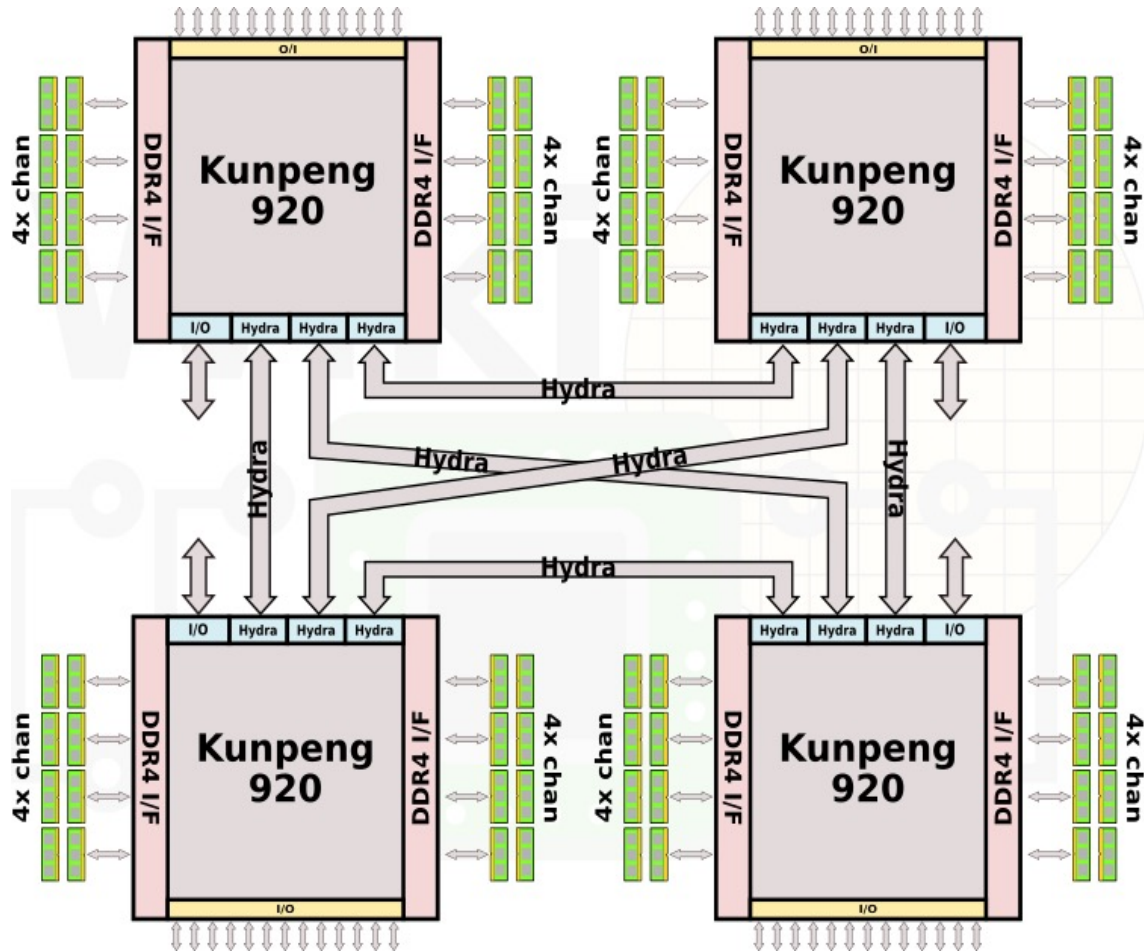
Hi1620 Specifications Overview



CPU core	Up to 64 ARMv 8.2 cores, 3.0 GHz, 48-bit physical address 4 issue OoO superscalar design 64 KB L1 I Cache and 64 KB L1 D cache 128-bit SIMD unit
L2 cache	512 KB private per core, 24 MB total
L3 cache	48 MB shared for all (1 MB/core), Partitioned
Memory	8-channel DDR4-2400/2666/2933/3200 16 ranks/channel, 1DPC and 2DPC configurations x4/x8 support ECC, SDDC, DDDC
PCIe	40 lanes of PCIe Gen4.0 16x
Integrated I/O	8 lanes of ETH, Combo MACs, supporting 2 x 100GE, 2 x 40GE, 8 x 25GE/10GE, 10 x GE, supporting SR-IOV RoCEv2/RoCEv1 x4 USB 3.0 x8 SAS 3.0 x2 SATA 3.0
Crypto engine	AES, DES/3DES, MD5, SHA1, SHA2, HMAC, CMAC Up to 100 Gbit/s
Compression	GZIP, LZS, LZ4 Up to 40 Gbit/s (compress)/100 Gbit/s (decompression)
RAID	RAID5/6, DIF, XOR, PQ acceleration
CCIX	Cache coherency interface for accelerator, like Xilinx FPGA World's 1st CCIX solution
Scale-up	Coherent SMP interface for 2P/4P 3*240Gbps bandwidth
Power	TDP ~150 W (48C 2.6 GHz)



The Huawei Kunpeng 920: multi-socket support



Next-Gen Kunpeng 930:

- higher-performance core
- SMT support
- Arm SVE (vector comp)
- expected in 2021



gone?...

1. Marvell ThunderX product family (dead?...)
2. Fujitsu A64FX Arm chip (in MACC...)
3. Neoverse N1 hyperscale reference design (...)
4. Ampere Altra Arm Processor (...)
5. Amazon Graviton (...)
6. Huawei HiSilicon Kunpeng 920 (...)

(2019)

Current ARMv8 & v9 HPCs: server-level competitors

(2022)

Technology Leadership
Arm ecosystem technology firsts

Fujitsu A64FX	Ampere Altra Max	AWS Graviton3	Alibaba Yitian 710
			
First 1TB/s Memory Bandwidth	First >100 cores per CPU (128)	First DDR5 PCIe Gen5.0 Memory and I/O chiplets	First >500 on SPEC CPU 2017 Integer Rate Source: https://www.sngp.org

© 2022 Arm

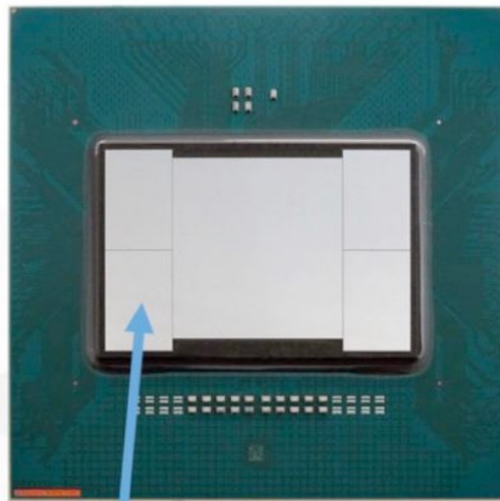
arm NEOVERSE

Fujitsu's A64FX ARM Chip

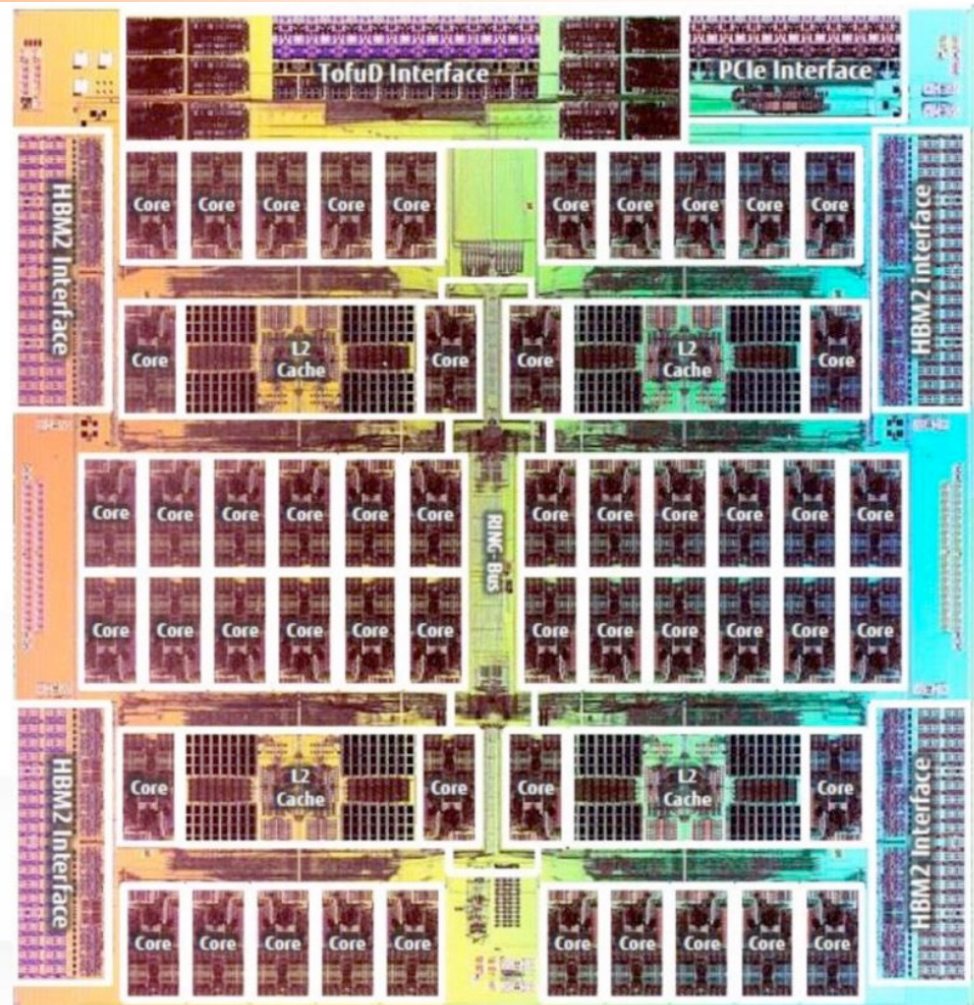
(#1 in June'20 TOP500)



- TSMC 7nm FinFET
- CoWoS technologies for HBM2



HBM2





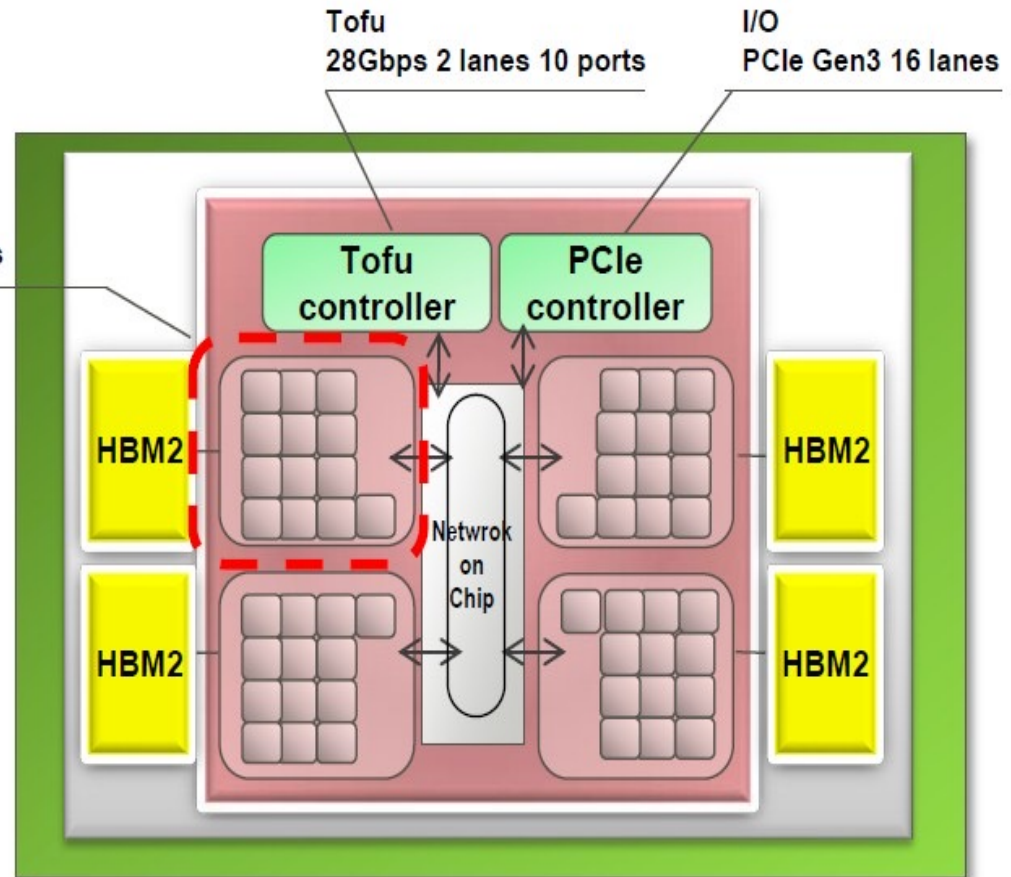
Fujitsu's A64FX Arm Chip: 48+4 cores

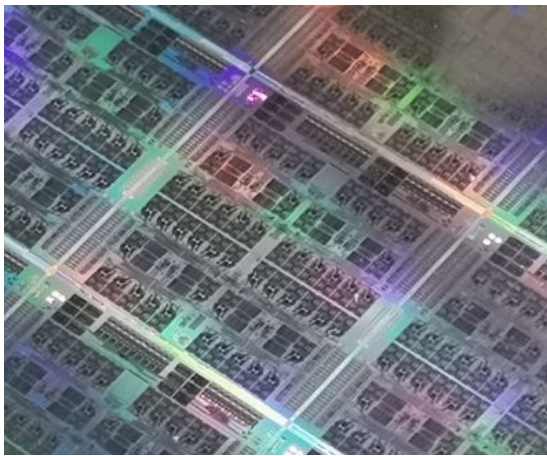


A64FX Arm

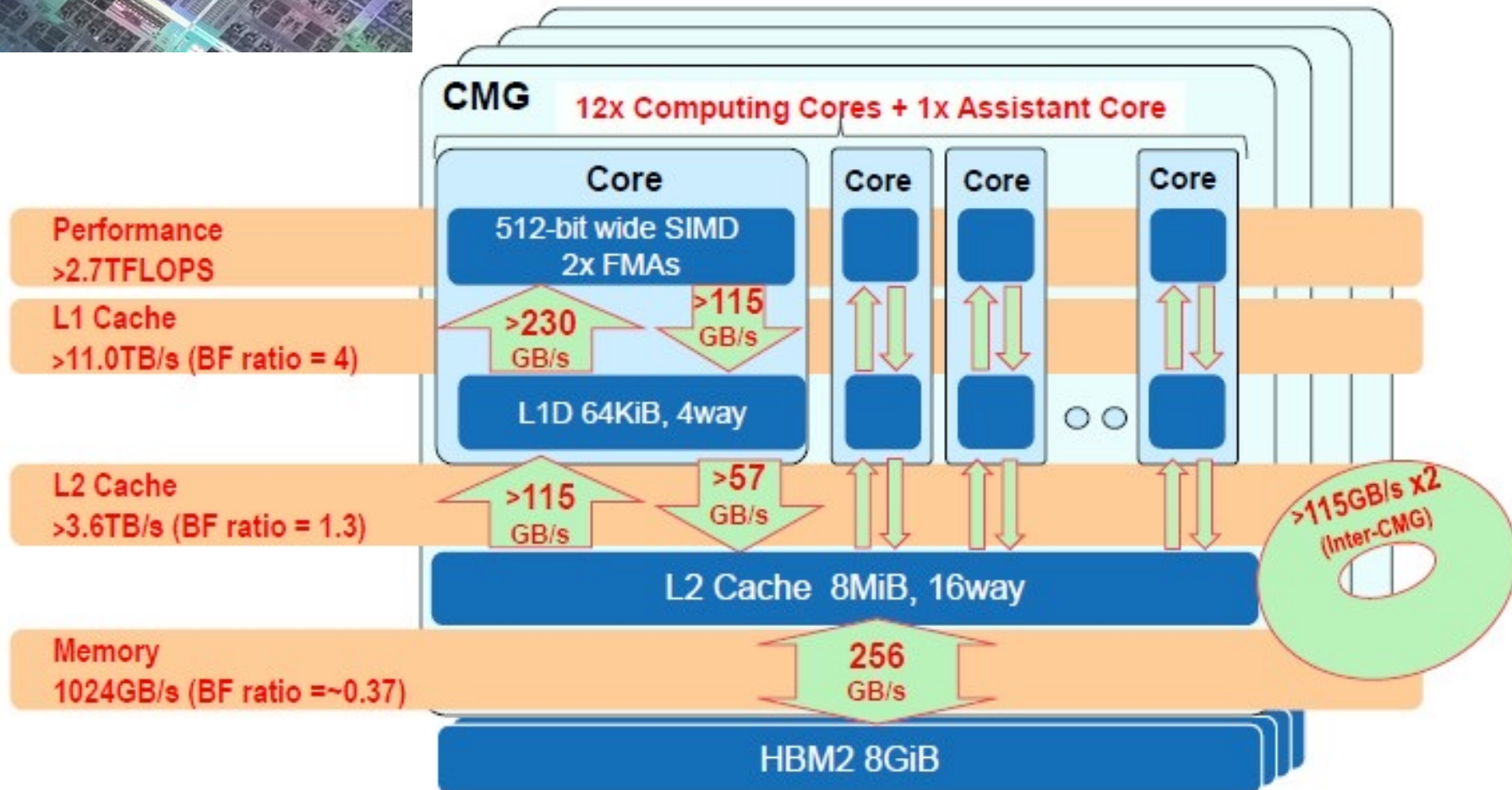
- Feature size: 7 nm
- Armv8.2-A spec with 512-bit SVE extensions
- HP math and a dot-product engine
- 4 core-memory groups
- NoC: a double ring bus
- cores in CMG linked by a crossbar to L2 cache & to HBM2 mem controller
- 8 MiB L2 cache; no L3 cache
- a Tofu-D controller on the die
- #1 TOP500 from Jun'20 to Nov'21 uses A64FX package

CMG specification
13 cores
L2\$ 8MiB
Mem 8GiB, 256GB/s





Block diagram of the A64FX chip





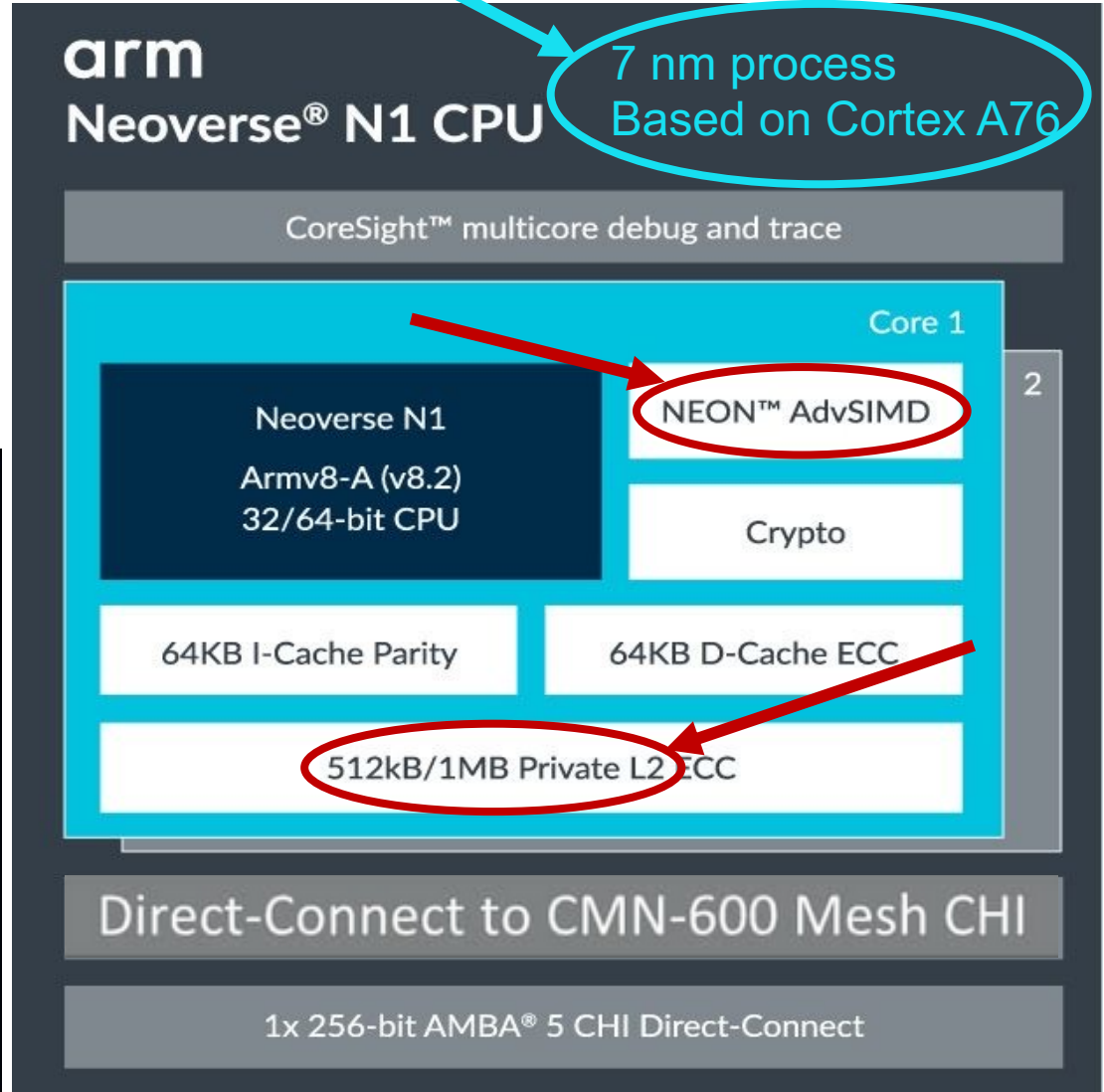
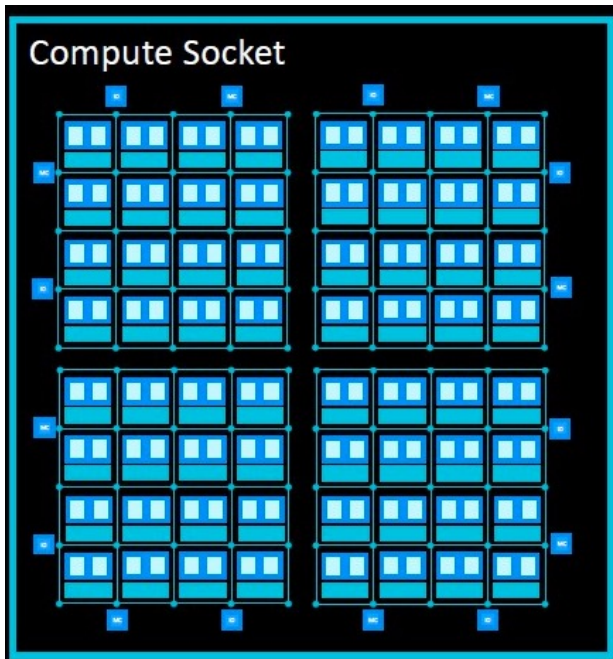
Arm Neoverse N1

(announced Feb'19)



Hyperscale
Datacenter

150W and beyond
64-128 core



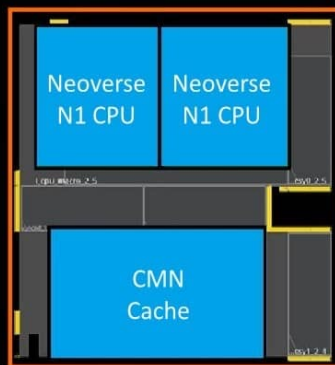


Arm Neoverse N1

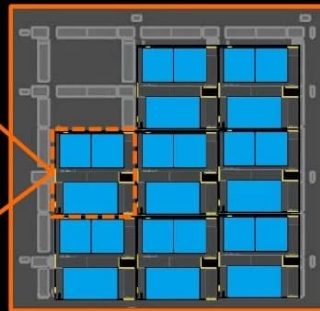
arm NEOVERSE

Building hyperscale compute in 7nm

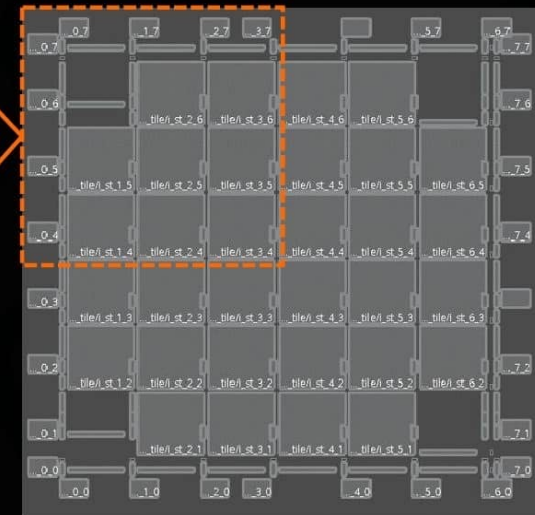
CPU Tile



Super Tile



Top-level Mesh



1.8-2.2GHz+
Coherent Mesh Network
CMN



Amazon Web Services (AWS): Graviton2 with Arm Neoverse N1 cores



AWS Graviton2

- 7 nm process
- up to 64 Arm Neoverse N1 cores
- 1 MiB L2 per core
- shared 32 MiB L3
- without SMT
- single-socket

AWS Designing a 32-Core Arm Neoverse N1 CPU for Cloud Servers

by [Anton Shilov](#) on December 2, 2019 1:00 PM EST

Posted in [Servers](#) [CPUs](#) [Arm](#) [Amazon](#) [AWS](#) [Neoverse N1](#)



Graviton1 Processor



First Arm-based processor in major cloud



Built on 64-bit ARM Neoverse cores with AWS-designed 16 nm silicon



Up to 16 vCPUs, 10 Gbps enhanced networking, 3.5 Gbps EBS bandwidth

Graviton2 Processor



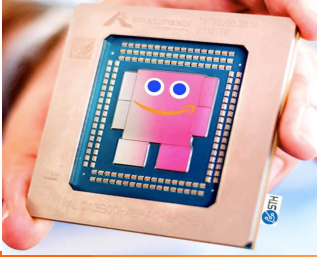
Built with 64-bit Arm Neoverse cores with AWS-designed 7 nm silicon process



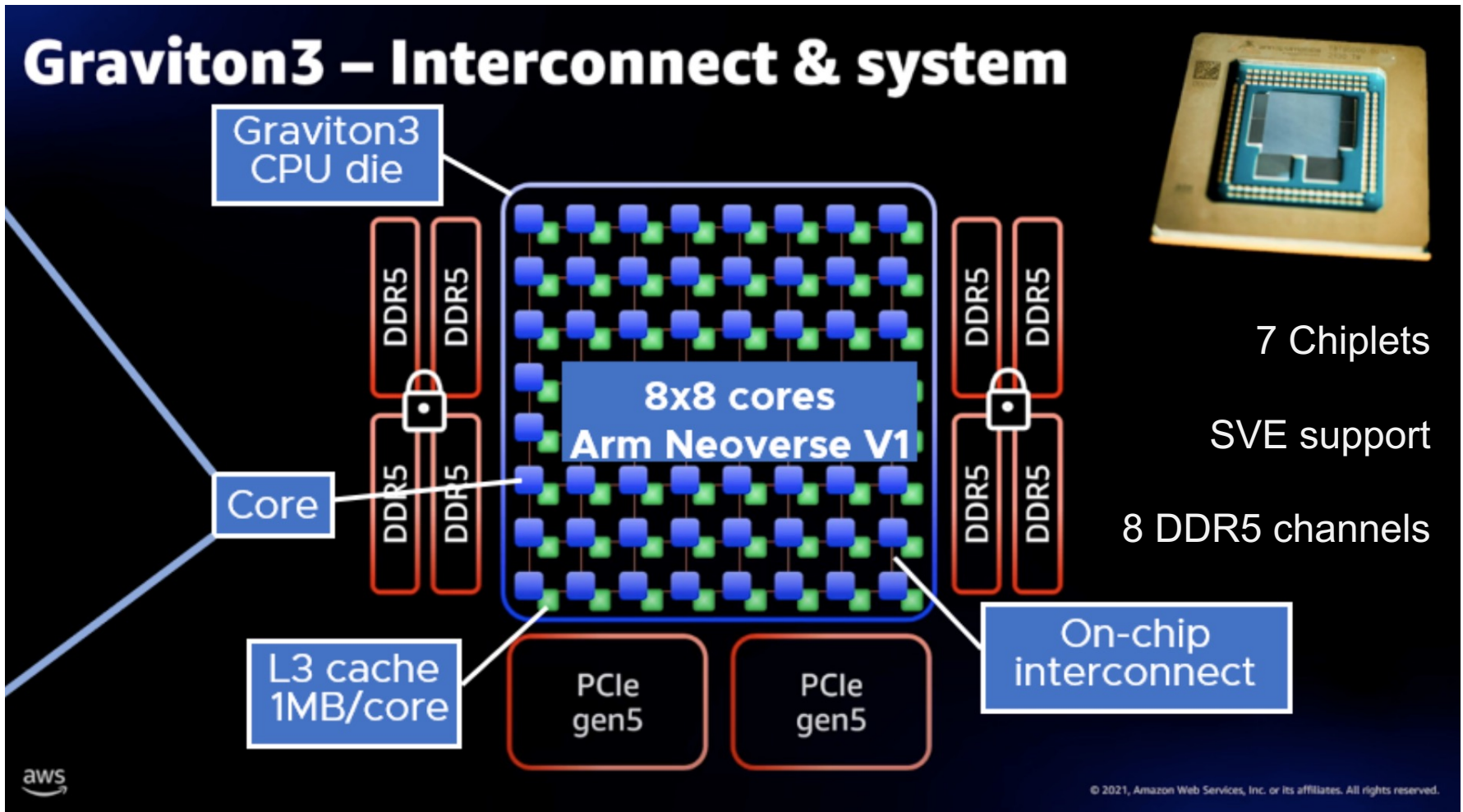
Up to 64 vCPUs, 25 Gbps enhanced networking, 18 Gbps EBS bandwidth



7x performance,
4x compute cores,
5x faster memory



AWS Graviton3 with Neoverse V1 cores





Alibaba Yitian 710



INVENT · EXPLORE · INSPIRE



Yitian

Born for Cloud

Alibaba's first general-purpose chip:
Yitian 710, compatible for different
application scenarios

ARMv9 Architecture

128 Arm cores

DDR5 Integral

PCIE5.0

YITIAN710

Manycore chips/packages: an overview on current servers



Key server chips/packages that addresses those issues:

- Intel: the Xeon Scalable family
- AMD: the Epyc Zen family
- ARM: key ARMv8 & v9 server-level competitors
 - Fujitsu A64FX Arm chip
 - Neoverse hyperscale reference design for Amazon Graviton
 - Alibaba Yitian 710
 - Huawei HiSilicon Kunpeng 920
- Sunway: the SX260x0 family
- **Cerebras: a Wafer Scale Engine**
- Apple (*not server*): the SoC approach (*no chiplets!*)



Cerebras: *a Wafer Scale Engine (WSE)* (Aug'19)



Cerebras Wafer Scale Engine (WSE):
the largest chip ever built

46,225 mm² chip

56x larger than the biggest GPU ever made

400,000 core

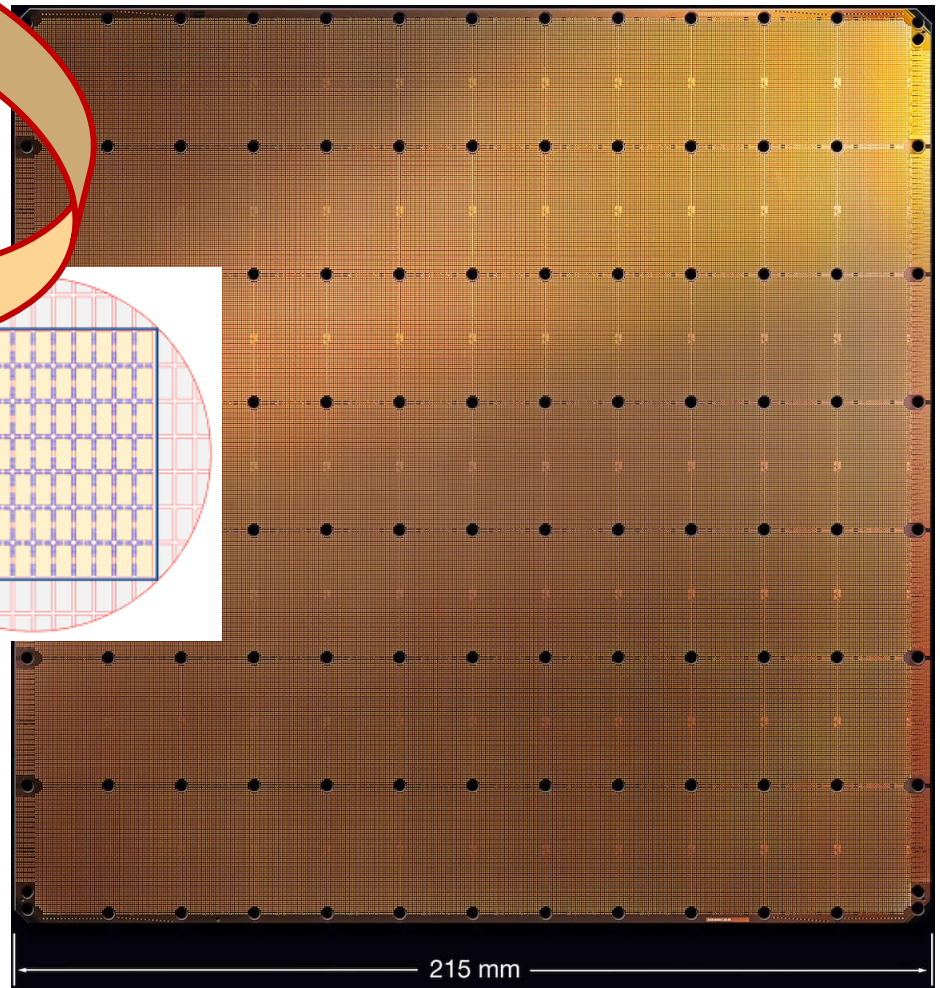
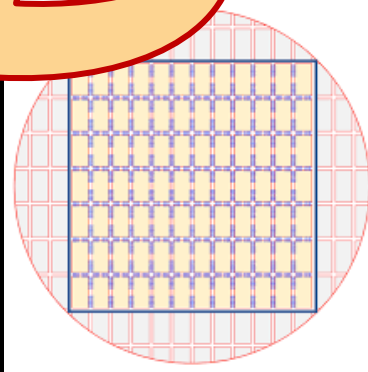
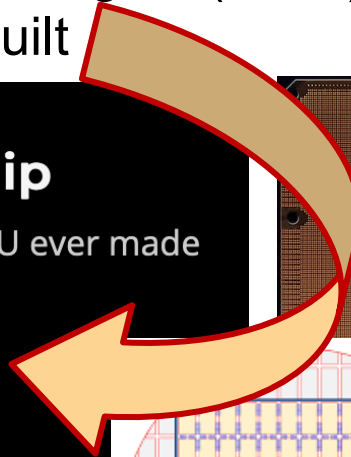
78x more cores

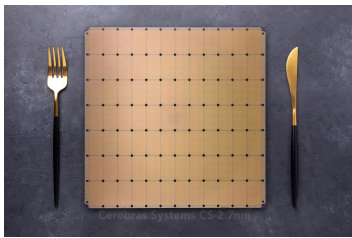
18 GB on-chip SRAM

3000x more on-chip memory

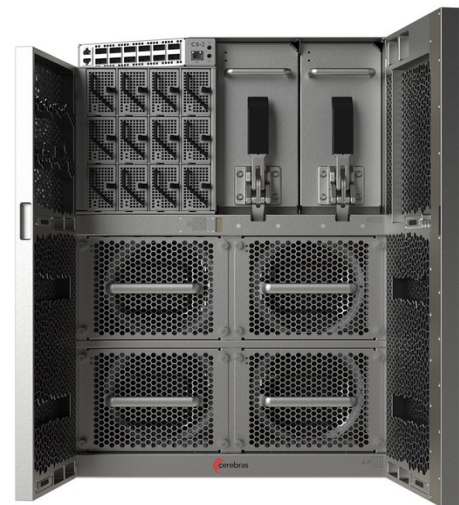
100 Pb/s interconnect

33,000x more bandwidth





Cerebras: 2nd Gen Wafer Scale Engine (WSE2) (Q3 2021)



	Cerebras Wafer Scale Engine 2	Cerebras Wafer Scale Engine	Nvidia A100
Process Node	TSMC 7nm	TSMC 16nm	TSMC 7nm N7
AI Cores	850,000	400,000	6,912 + 432
Die Size	46,255 mm ²	46,255 mm ²	826 mm ²
Transistors	2.6 Trillion	1.2 Trillion	54 Billion
On-Chip SRAM Memory	40 GB	18 GB	40 MB
Memory Bandwidth	20 PB/s	9 PB/s	1,555 GB/s
Fabric Bandwidth	220 Pb/s	100 Pb/s	600 GB/s
Power Consumption (System/Chip)	20kW / 15kW	20kW / 15kW	250W (PCIe) / 400W (SXM)

Manycore chips/packages: an overview

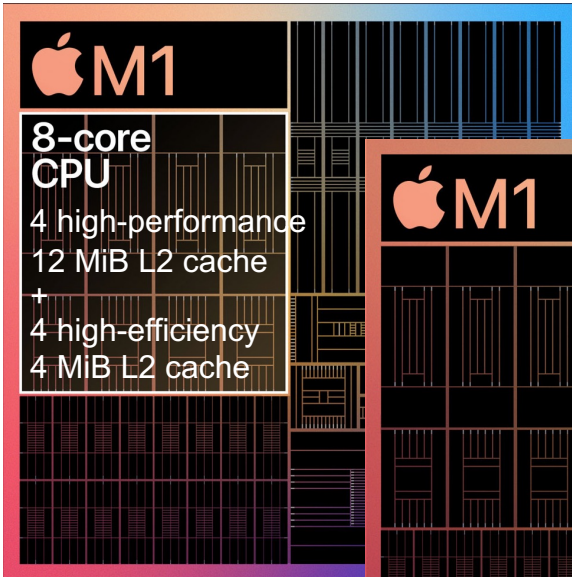


Key server chips/packages that addresses those issues:

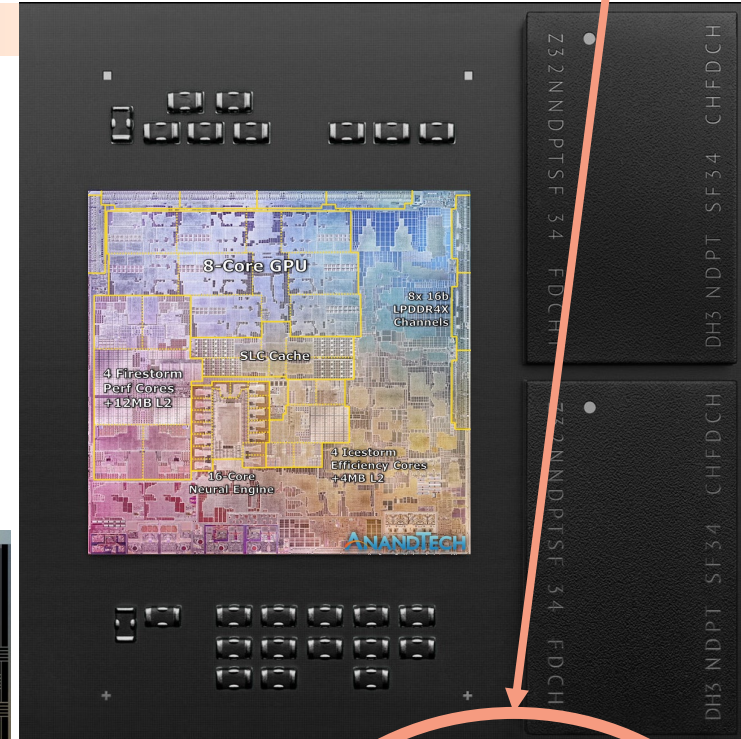
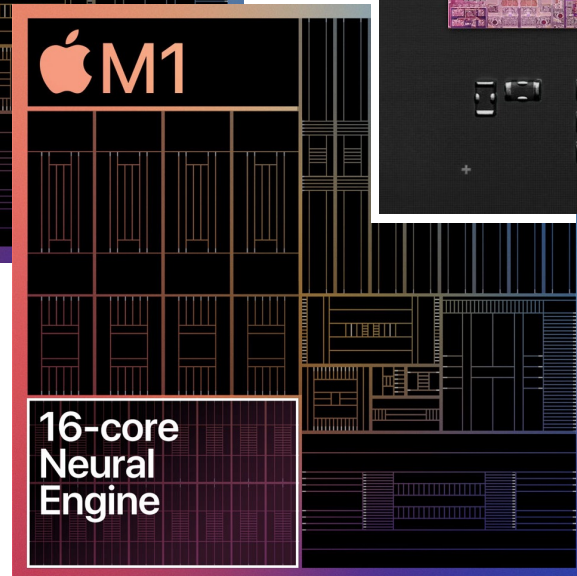
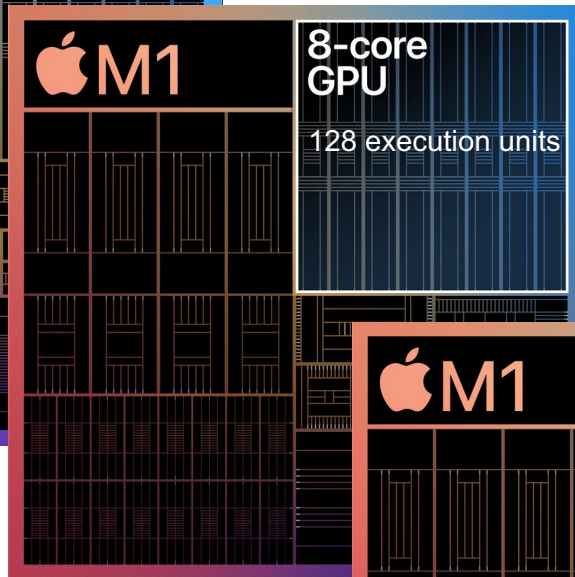
- Intel: the Xeon Scalable family
- AMD: the Epyc Zen family
- ARM: key ARMv8 & v9 server-level competitors
 - Fujitsu A64FX Arm chip
 - Neoverse hyperscale reference design for Amazon Graviton
 - Alibaba Yitian 710
 - Huawei HiSilicon Kunpeng 920
- Sunway: the SX260x0 family
- Cerebras: a Wafer Scale Engine
- **Apple** *(not server)*: **the SoC approach** *(no chiplets!)*



Apple M1 SoC (announced Nov'20)

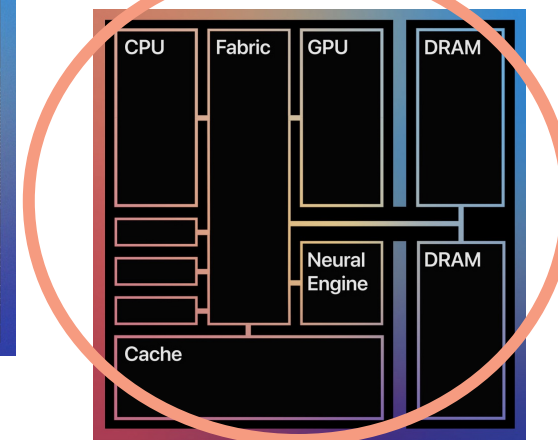


5-nanometer
process



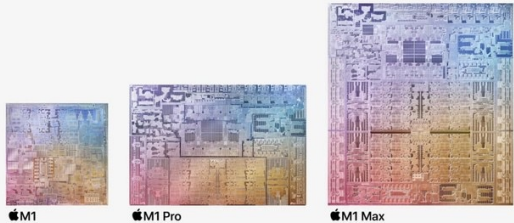
Apple claims SoC is better:

- shorter latencies
- better silicon process (5 nm)
- better wafer fabrication (increased yield)



M1 successors: M1 Pro & M1 Max

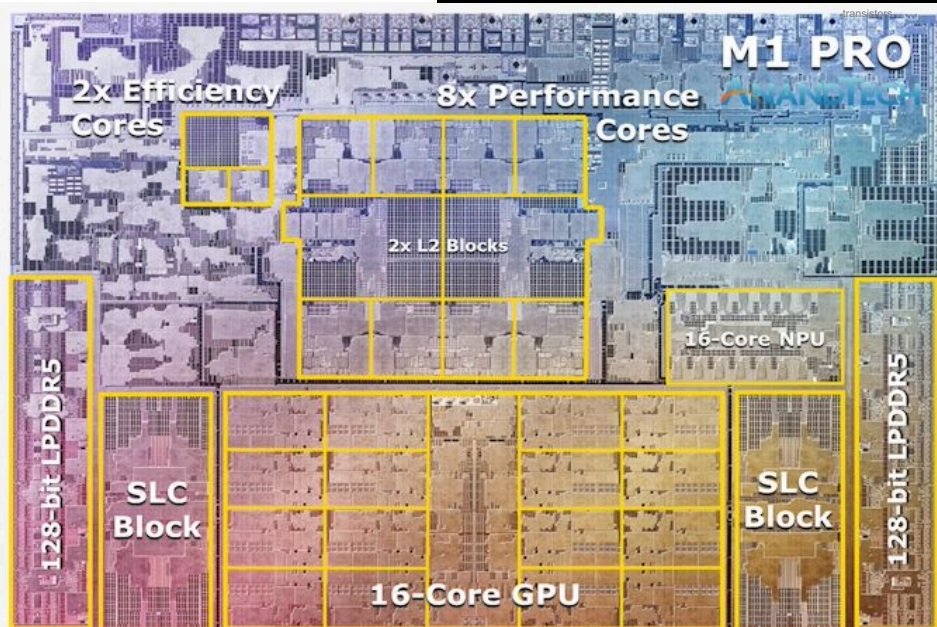
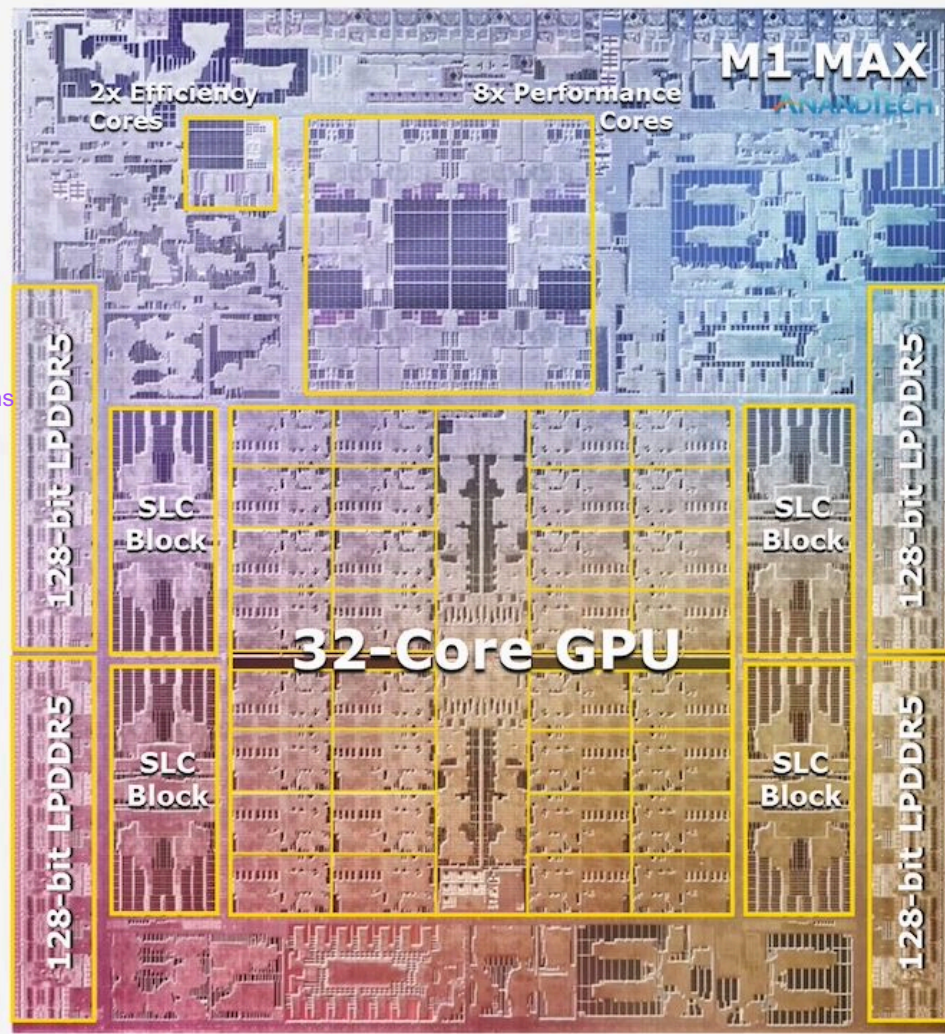
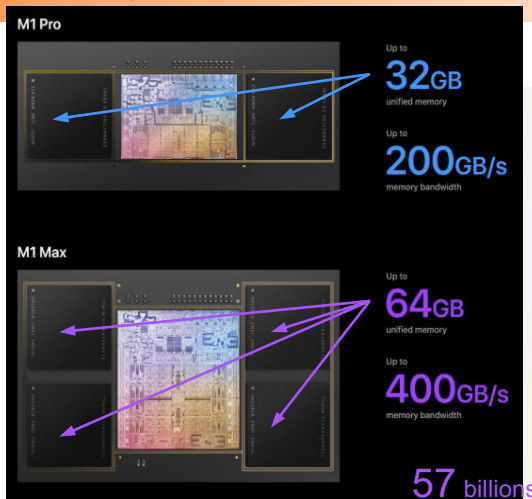
(announced Oct'21)



<https://www.anandtech.com/show/17019/apple-announced-m1-pro-m1-max-giant-new-socs-with-allout-performance>

Mi Pro
256-bit LPDDR5-6400
⇒ $32B * 6.4GHz$
≈ 200 GB/s

Mi Max
512-bit LPDDR5-6400
⇒ $64B * 6.4GHz$
≈ 400 GB/s



M1 successors: M1 Ultra (announced Mar'22)

<https://www.anandtech.com/show/17306/apple-announces-m1-ultra-combining-two-m1-maxes-for-even-more-performance>

The graphic displays the specifications for the Apple M1 Ultra chip. At the center is the 'Apple M1 ULTRA' logo. Surrounding it are various performance metrics: 'ProRes' for encode and decode, 'Thunderbolt 4' with a lightning bolt icon, '5 nm process', '114 billion Transistors', '2.5TB/s' for interprocessor bandwidth using a 'Silicon Interposer', '800GB/s' for memory bandwidth, 'UltraFusion architecture' with a die image, and 'Up to 128GB' of unified memory. On the left, it lists '20-core CPU', '64-core GPU', and a '32-core Neural Engine' capable of '22 trillion operations per second'. At the bottom, it mentions 'Secure Enclave' and 'Industry-leading performance per watt'. A large red arrow points from the bottom left towards the '32-core Neural Engine' section. On the right, a circular inset shows a die image with two M1 Max chips highlighted by pink ovals. Arrows from these ovals point to the 'Up to 128GB' unified memory specification, indicating that the M1 Ultra's memory is composed of two M1 Max chips.

ProRes
Encode and decode

Thunderbolt 4

5 nm process

114 billion Transistors

2.5TB/s
interprocessor bandwidth
Silicon Interposer with

800GB/s
Memory bandwidth

Apple M1 ULTRA

UltraFusion architecture

Up to 128GB
unified memory

20-core CPU

64-core GPU

32-core Neural Engine
22 trillion operations per second

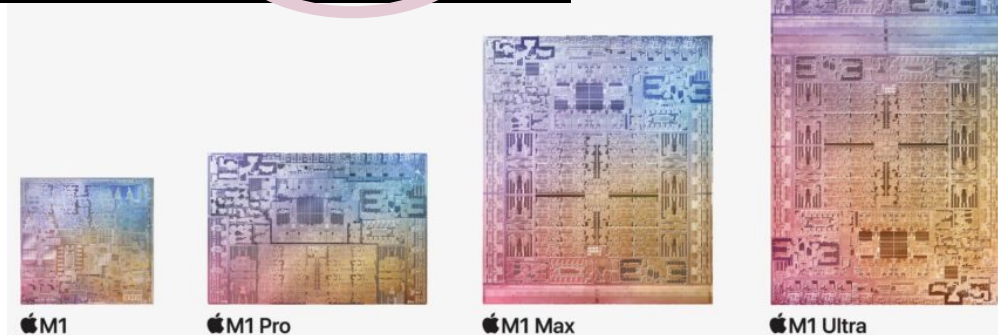
Secure Enclave

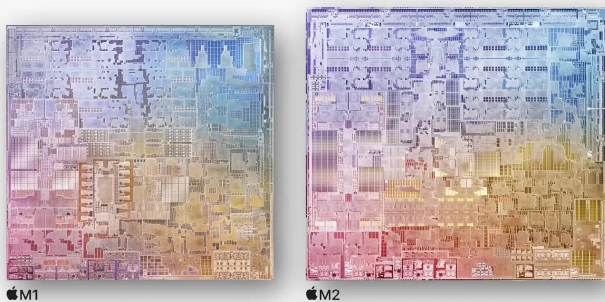
Industry-leading performance per watt

M1 Ultra:
2 M1 Max



APereira, AJProença, SAC, 2025

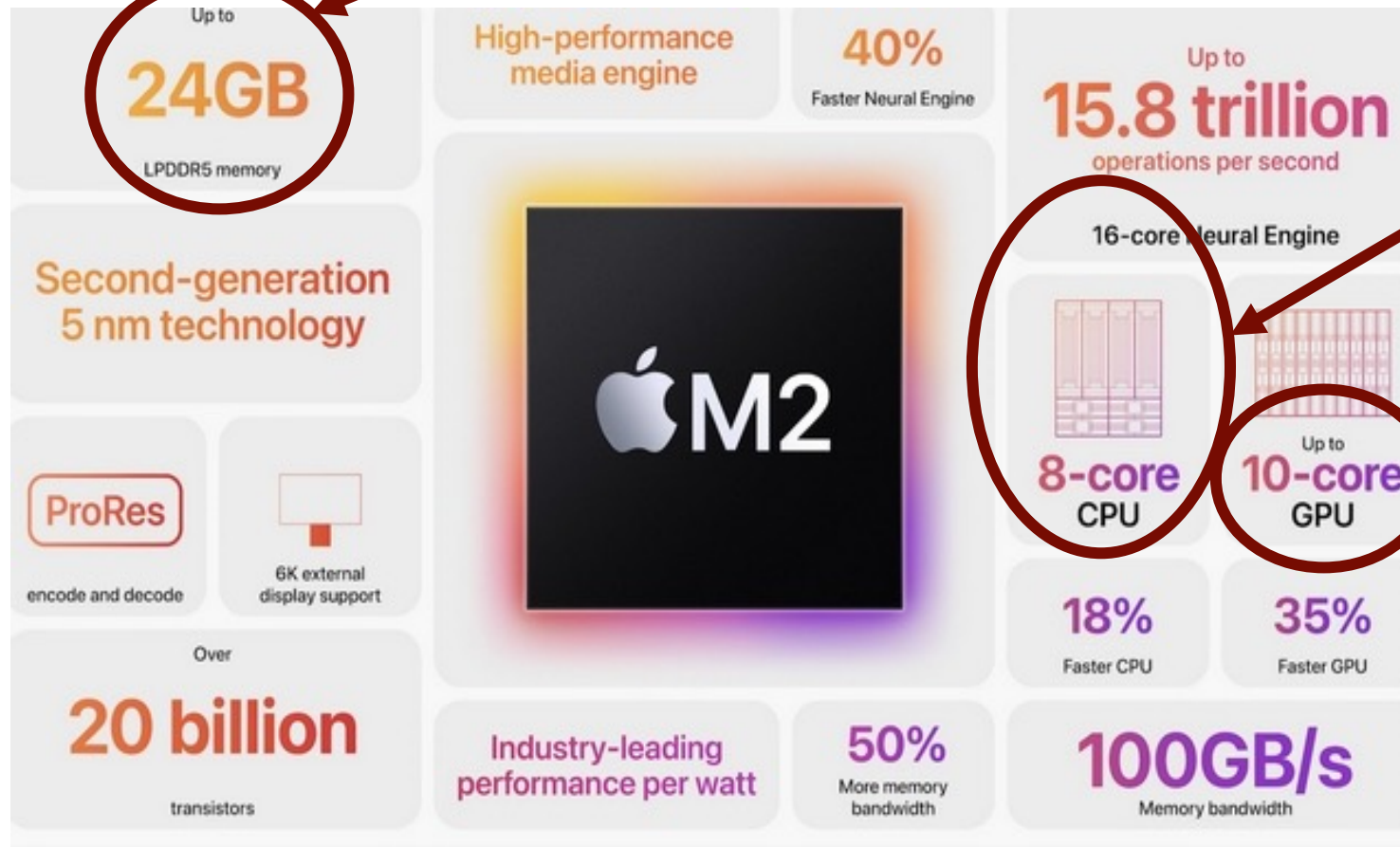




M1 successors: a faster M2

(announced Jun'22)

Up from 16 GiB LPDDR4 in M1



Same as M1

+2 than M1



M2 successors: M3 generation

(launched Fall'23)



Preliminary data



Thunderbolt 4

Up to
20%

faster CPU

Up to
30%

faster GPU

Industry-leading
performance
per watt

Up to
96GB

LPDDR5 memory

Same in
M3 Max

Over
67 billion
transistors

16-core
**Neural
Engine**

15.8 trillion ops/s

40%

Faster Neural Engine

High-performance
media engine with ProRes

**M2
MAX**



12-core
CPU



Up to
38-core
GPU

16-core in
M3 Max

40-core in
M3 Max

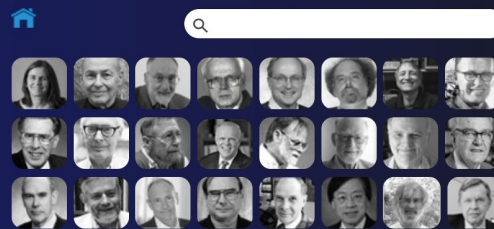
400GB/s
Memory bandwidth

Second-generation
5 nm technology

3nm

Smaller transistors

M3 Ultra = 2x M3 Max



The Turing Award



Turing Award

From Wikipedia, the free encyclopedia



The **ACM A. M. Turing Award** is an annual prize given by the [Association for Computing Machinery](#) (ACM) for contributions of lasting and major technical importance to [computer science](#).^[2] It is generally recognized as the highest distinction in [computer science](#) and is colloquially known as or often referred to as the "[Nobel Prize of Computing](#)".^{[3][4][5][6]}

The award is named after [Alan Turing](#), who was a British mathematician and [reader](#) in mathematics at the [University of Manchester](#).

2017

John L. Hennessy



David Patterson



turing lecture

Nobel equivalent in Computer Science

DOI:10.1145/3282307

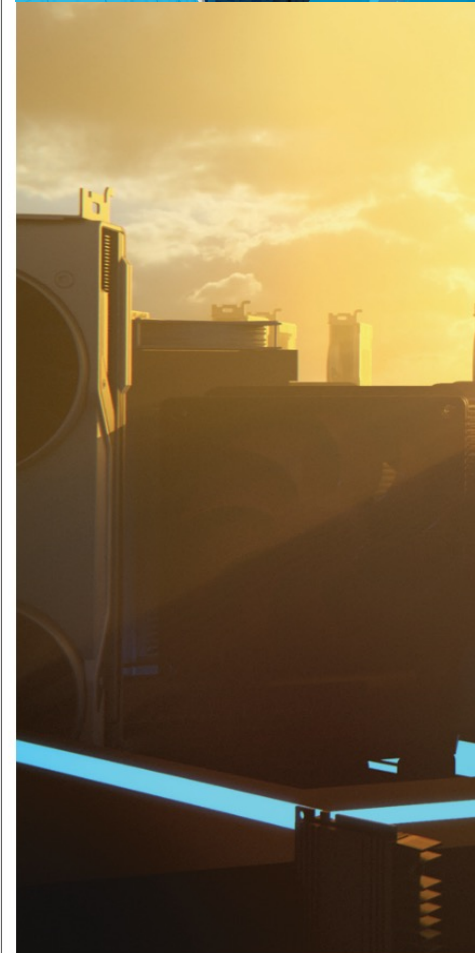
Innovations like domain-specific hardware, enhanced security, open instruction sets, and agile chip development will lead the way.

BY JOHN L. HENNESSY AND DAVID A. PATTERSON

A New Golden Age for Computer Architecture

WE BEGAN OUR Turing Lecture June 4, 2018¹¹ with a review of computer architecture since the 1960s. In addition to that review, here, we highlight current challenges and identify future opportunities, projecting another golden age for the field of computer architecture in the next decade, much like the 1980s when we did the research that led to our award, delivering gains in cost, energy, and security, as well as performance.

"Those who cannot remember the past are condemned to repeat it."
—George Santayana, 1905



engineers, including ACM A.M. Turing Award laureate Fred Brooks, Jr., thought they could create a single ISA that would efficiently unify all four of these ISA bases.

They needed a technical solution

And now?